The long-term effects of the Tusi system on firm entry: The role of Targeted Poverty Alleviation

Abstract

Understanding the persistent impact of historical institutions is crucial for promoting regional coordinated development and common prosperity. Combining historical Tusi records with firm entry data from 183 counties in Sichuan province, we examine the long-term effects of the Tusi system on contemporary firm entry. We find that the Tusi legacy significantly inhibits contemporary firm entry. Quantitatively, regions with a Tusi legacy experience a 24% lower level of modern firm entry compared to their conventional counterparts. Mechanism analysis reveals that this suppression operates through two dimensions: "soft factor constraints" (stifling modern finance and human capital) and "hard input barriers" (underdeveloped infrastructure). Notably, the inhibitory effect intensifies with the historical administrative rank of the Tusi office, whereas the mere geographical quantity of such offices shows no significant scale effect. Furthermore, employing a triple-difference (DDD) framework, this study evaluates the offsetting impact of Targeted Poverty Alleviation (TPA). The results show that TPA, as a robust modern intervention, significantly mitigates the inhibitory effect of Tusi legacy on firm entry. By identifying the long-term crowding-out effect of this predatory institution, this paper provides empirical evidence for understanding historical barriers within Chinese-style modernization. Moreover, the evidence confirms that modern governance can effectively repair historical institutional scars, offering Chinese insights for underdeveloped regions worldwide to break path dependency and achieve synergistic transformations of institutions and markets.

Keywords: Tusi system; Firm entry; State capacity; Path dependence; Targeted Poverty Alleviation

1. Introduction

The divergence in contemporary regional economic development is not merely a product of current endowments but is profoundly rooted in historical institutional legacies. In recent years, the persistence of historical determinants has emerged as a cornerstone of interdisciplinary research spanning economics and history (Acemoglu et al., 2001,2015; Iyer, 2010; Johnson & Koyama, 2017; Mattingly, 2017). Leveraging quasi-natural experiments such as geographic regression discontinuity designs (GRDD), existing literature has documented the enduring impact of colonial-era systems, like the Peruvian Mita (Dell, 2010), and historical industrial legacies on contemporary local culture (Huggins et al., 2021) on modern economic disparities. However, to decipher the transmission mechanisms of these institutional shocks, it is imperative to scrutinize the micro-behavior of firms—the foundational actors of a market economy. While emerging studies have explored how historical factors, such as the imperial examination system (Liu et al., 2024) and merchant guild cultures (Wang et al., 2025), shape corporate innovation and financing, empirical evidence regarding firm entry remains sparse. Specifically, there is a significant research lacuna at the intra-provincial level concerning how historical administrative institutions anchor contemporary entrepreneurial vitality and delineate market access barriers.

The number of firm entries is not only a key indicator of a city's economic development but also a micro-level window into its institutional environment. As a primary driver of market transformation and resource allocation (Asturias et al., 2023), the decision to enter a market depends not only on economic factors like costs but also on "invisible barriers" within the institutional framework, such as informal institutions (Sun et al., 2024).However, while existing studies have pointed out the potential impact of historical legacies on firm behavior, this influence is often viewed as static. In reality, to accurately identify the actual force of historical legacies, we must examine how micro-actors respond dynamically to modern policy interventions. Therefore, focusing on firm entry can do more than just reveal the long-term impact of historical institutions; it provides micro-level evidence of how contemporary governance policies can overcome historical inertia.

China's Targeted Poverty Alleviation (TPA) campaign—a landmark endeavor in global poverty reduction—provides an exceptional quasi-natural experiment to observe dynamic institutional responses. Since 2015, the campaign has expanded beyond its primary mandate of eradicating absolute poverty. Drawing on Banerjee & Duflo's (2011) insights, the impoverished do not fundamentally diverge from the general population in terms of intrinsic preferences or rationality; rather, their predicament is underpinned by a chronic deficit of reliable information and a heightened susceptibility to cognitive fluctuations. Consequently, bridging the gap between theoretical frameworks and practical interventions, China's poverty alleviation strategy has moved beyond traditional paradigms. Through massive infrastructure investment, industrial subsidies, and social mobilization—specifically the "Alleviating poverty by enhancing both aspiration and intelligence" initiative—the policy has sought to dismantle entrenched poverty traps by addressing the informational and psychological bottlenecks that perpetuate deprivation. By mitigating the psychological constraints and behavioral biases associated with chronic deprivation (Duflo, 2012), the TPA has fundamentally reshaped the governance architectures and business environments of underdeveloped regions. Following the total elimination of absolute poverty and the removal of all

832 designated counties from the poverty list by 2020, extant literature has documented its multifaceted impacts. At the micro level, the campaign has bolstered rural household incomes (Li et al., 2025) and subjective well-being (Zhou et al., 2023); at the macro level, it has catalyzed industrial diversification and stimulated economic growth (Liu et al., 2025; Wang et al., 2025). Crucially, these targeted interventions have been shown to foster entrepreneurship among rural residents and vulnerable groups (Zhang et al., 2025). Therefore, treating the TPA as a potent modern policy intervention allows for a rigorous scrutiny of whether proactive state governance can effectively counteract the historical inertia rooted in legacy institutions (Robinson& Acemoglu, 2012), thereby lowering entry barriers and revitalizing market dynamism.

Sichuan Province in China is identified as an ideal "natural laboratory" for this study to empirically examine the persistent effects of historical institutions. This selection is justified, first, by the remarkable continuity of Chinese administrative evolution and its rigorous tradition of official record-keeping, which provide a highly stable and authoritative empirical foundation for long-term quantitative analysis. More importantly, under the framework of a centralized state, the geographic demarcation of the Longmen Mountains fostered a unique "dual governance structure" within a single province that persisted for over six centuries: the eastern agricultural basin remained under direct central administration (Commandery-county System[1]), while the western plateau was governed primarily through the Tusi system[2]. Distinct from the standard bureaucratic governance, the Tusi system granted local leaders hereditary rights, fiscal autonomy, military authority, and cultural preservation powers. This intra-provincial institutional diversity allows for the scrutiny of institutional variance at a much more granular scale. By effectively neutralizing the confounding factors typically encountered in cross-national research, this setting minimizes potential biases and enhances the efficiency and quality of causal identification (Feng, 2019). Consequently, Sichuan

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[1] The commandery-county system *(Junxian* system) represents one of the most pivotal administrative frameworks in ancient China. Originating during the Spring and Autumn and Warring States periods and reaching its zenith under the Qin and Han dynasties, this two-tier local governance structure functioned as a cornerstone of the centralized state. Under this system, the central government exercised vertical control over local jurisdictions through a formal bureaucracy, where local officials were directly appointed and removed by the Emperor based on fixed tenures. By bringing local administration under direct central oversight, this system effectively strengthened political centralization and facilitated national unification.

[2] The Tusi system was an institutional framework for local governance deployed by the central dynasties of the Yuan, Ming, and Qing to manage ethnic minority regions in Southwest and Northwest China. Characterized by indirect rule, the system involved the formal investiture of hereditary local chieftains (*Tusi*), who exercised autonomous administrative, military, and land jurisdiction. In exchange, these chieftains fulfilled obligations to the central state, including paying tribute and providing military levies, thereby establishing a governance structure described as "ruling barbarians through barbarians". Tracing its origins to the *Jimi* (loose rein) policies of the Qin and Han and evolving through the Tang and Song, the Tusi system was formally institutionalized during the Yuan dynasty. The Ming dynasty initiated the "*Gaitu Guiliu*" reform—a process of bureaucratization that gradually replaced hereditary chieftains with centrally appointed officials (*liuguan*)—until the system's final dissolution following the land reforms of the 1950s. While the Tusi system initially enhanced frontier governance and facilitated inter-ethnic economic integration, it eventually hindered social progress as entrenched local strongholds and predatory exactions by chieftains exacerbated tensions between the periphery and the center.

provides a precise empirical environment to estimate the causal impact of historical administrative legacies on contemporary firm entry.

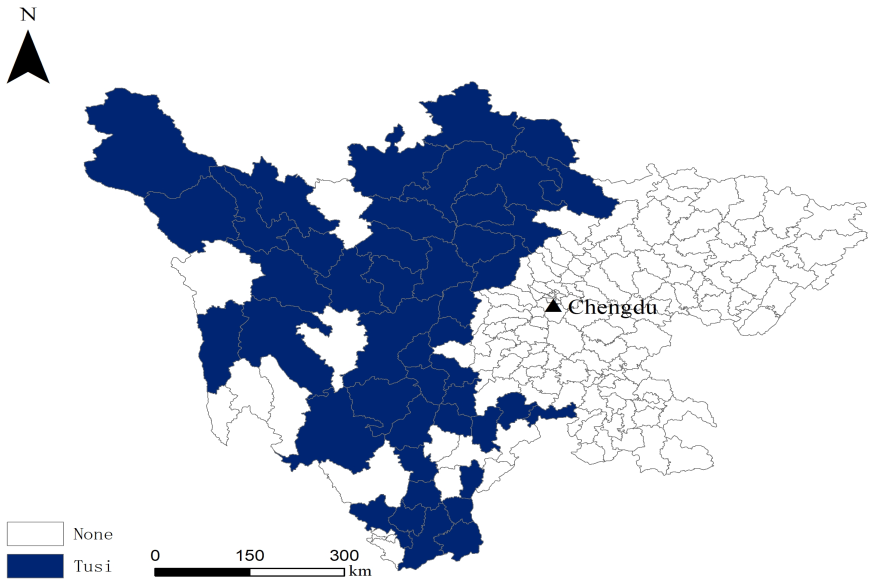

Fig.1 The historical implementation of the Tusi system in Sichuan

Notes: This map illustrates the historical geographic distribution of the Tusi system in Sichuan Province. Regions that implemented the Tusi system are highlighted in dark tones, while regions without the Tusi system are shown in white. (Map Approval Number: GS (2024) 0650)

Against this backdrop, this study investigates the persistent impact of the Tusi system on contemporary firm entry and uncovers its underlying transmission mechanisms. Grounded in State Capacity Theory and Path Dependence Theory, we develop a theoretical framework to hypothesize how historical governance legacies shape modern micro-level entrepreneurial activity. For our empirical strategy, we utilize a unique, hand-collected dataset of historical Tusi seats across 183 counties in Sichuan Province, which is then matched with modern firm-level entry data. This effort results in a comprehensive panel dataset comprising 2379 observations spanning from 2011 to 2023. Moving beyond a simple estimation of the long-term deterrent effects and their heterogeneities, this paper further conceptualizes the Targeted Poverty Alleviation campaign as a potent modern institutional shock. This allows us to scrutinize how contemporary governance interventions counteract and reshape historical institutional legacies.

This paper contributes to the existing literature in three primary dimensions. First, regarding the research perspective, this study extends the scope of historical institutionalism. Unlike extant literature that primarily focuses on the impact of historical institutions on macro-economic outcomes, such as regional wealth gaps (Dell, 2010), we shift the focus to the micro-level dynamics of firm entry. This approach not only provides direct empirical evidence of how historical legacies shape the decisions of micro-agents but also enriches the academic discourse on the deep-rooted origins of market entry barriers.

Second, in terms of identification strategy and data construction, we leverage the “natural laboratory” of Sichuan’s dual governance structure to enhance causal inference. While prior studies on the long-term effects of institutions on entrepreneurship often rely on cross-national or inter-regional comparisons (Glaeser, 2002; Redding, 2008), our intra-provincial analysis effectively controls for unobserved heterogeneities in geography, culture, and macro-policies. Furthermore, the unique dataset of historical Tusi seats, hand-collected from primary archival sources, offers a rare and rigorous empirical foundation for quantifying institutional transitions in Southwest China.

Finally, by examining modern policy shocks, this study validates the effectiveness of the “bolstering aspirations before alleviating poverty” approach. Our empirical analysis demonstrates how institutional reforms can trigger profound ideological and cultural shifts, which in turn reshape economic outcomes. This not only uncovers the interactive logic between modern governance and historical legacies but also confirms that potent contemporary policies can indeed break historical path dependence. Such a finding offers vital policy insights for regions burdened by historical constraints to optimize their business environments and foster endogenous growth.

The remainder of this paper is structured as follows. Section 2 establishes the historical background and theoretical framework. Section 3 details the data and empirical strategy. Section 4 presents the main results, while Section 5 explores the mechanisms and heterogeneities. Section 6 provides an extended analysis. Section 7 concludes.

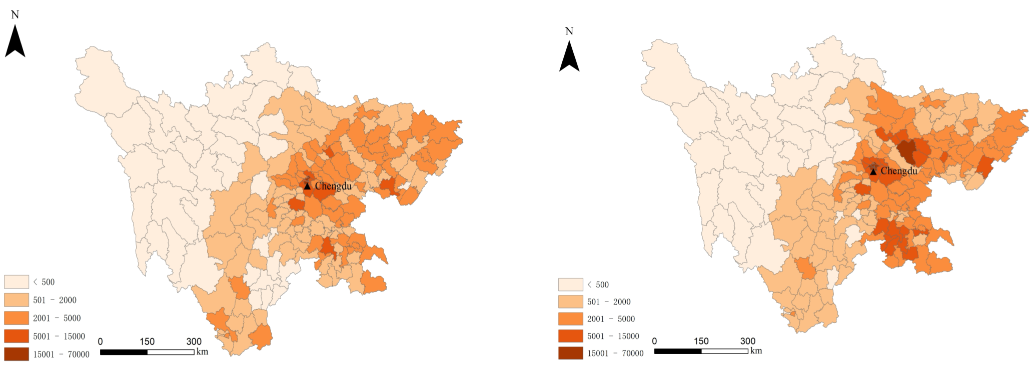

Fig.2 Comparison of new firm entries

Notes: These two maps present a comparison of the number of newly entered firms in Sichuan Province between 2011 and 2023. The left map shows the distribution for 2011, while the right map corresponds to 2023. The figures indicate that non-Tusi areas exhibit a more significant deepening of color, reflecting a higher volume of firm entries. (Map Approval Number: GS (2024) 0650)

2. Historical background, Policy context and theoretical framework

2.1 Historical background

The Tusi system was a distinctive frontier governance institution developed by imperial China to administer ethnically diverse border regions. Under this system, the imperial court recognized local hereditary chieftains as official administrators and granted them titles, seals, and limited autonomy in exchange for loyalty, tribute, and military service (Herman, 2007). Through this arrangement, the central government exercised indirect rule over frontier societies. Meanwhile, local leaders were allowed to maintain authority over their communities and customary institutions (Gong, 2012). The institutional foundations of the Tusi system emerged during the Yuan dynasty (1271–1368) and were further formalized and expanded during the Ming dynasty (1368–1644) (Gong, 2012). During the Qing dynasty (1644–1911), the system continued to operate but gradually declined as the imperial government attempted to strengthen direct administrative control over frontier territories. Although elements of hereditary chieftain authority survived into the Republican period, the remaining Tusi jurisdictions were eventually abolished in the 1950s when the new government incorporated these regions into the modern administrative system of the People's Republic of China (Yang, 2009).

Geographically, the Tusi system was concentrated primarily in southwestern China, especially Sichuan, Yunnan, and Guizhou provinces. These regions featured rugged mountains and fragmented transportation networks, which made direct imperial governance difficult (Swope, 2009). Southwestern China was also home to extraordinary ethnic diversity, with numerous groups maintaining distinct languages, customs, and social structures (Harrell, 2001). Under such conditions, governing through recognized local leaders provided a pragmatic solution that enabled the imperial state to extend political authority without establishing a costly bureaucratic apparatus in remote areas. In Sichuan, the Tusi system was particularly prominent in three regions: the Liangshan Yi region, the Ganzi Tibetan areas, and the Aba Qiang territories. In these areas, hereditary chieftains exercised significant authority over land and local populations. The imperial court formally recognized these leaders through official appointments and symbolic regalia while defining the boundaries of their jurisdictions (Millward, 1998). In return, Tusi rulers were expected to submit tribute, maintain local order, and provide military assistance when mobilized by the state. This arrangement created a form of semi-autonomous frontier governance in which imperial sovereignty coexisted with substantial local autonomy (Herman, 2007).

Institutionally, Tusi leaders possessed broad administrative and judicial authority within their territories. The Tusi yamen was the central institution of local governance. It served as an administrative office, a judicial court, and a ceremonial center simultaneously (Herman, 2007). Many disputes were resolved according to customary ethnic laws rather than the imperial legal code.

These local legal practices were widely accepted within communities and helped maintain social order in frontier societies (Harrell, 2001).

The economic consequences of the Tusi system were complex. Many Tusi leaders controlled strategic resources and trade routes through monopolies and taxation. Commodities such as Zigong salt, Ya'an tea, and mineral resources from Liangshan were often regulated by local chieftains, allowing them to extract significant revenues from both local populations and outside merchants. The system also intersected with regional trade networks linking the Tibetan Plateau, southwestern China, and interior markets (Perdue, 2005). At the same time, restrictions imposed within Tusi territories sometimes limited the integration of frontier markets into the broader imperial economy. Despite official prohibitions on Han migration and certain forms of commerce, informal trade networks developed along mountain routes (Yang, 2009). Local traders exchanged furs, medicinal herbs, textiles, and metal goods through cross-border markets and smuggling channels, creating alternative commercial systems partly outside formal state regulation.

From the late Ming period onward, the imperial state increasingly attempted to replace hereditary chieftain authority with direct bureaucratic administration through the policy known as "*gaitu guiliu*". This reform involved abolishing native chieftain rule and replacing it with centrally appointed officials integrated into the regular county–prefecture administrative hierarchy (Gong, 2012). Large-scale implementation occurred particularly during the Yongzheng reign (1723–1735), when many Tusi jurisdictions in southwest China were converted into regular administrative units (Herman, 2007).

Scholars have offered different interpretations of the historical significance of the Tusi system. Some view it as a pragmatic frontier governance strategy that enabled the imperial state to rule diverse populations with relatively low administrative costs (Herman, 2007). Others emphasize its role in a broader process of imperial expansion and incorporation of frontier societies into the Chinese state (Hostetler, 2001). Despite these different interpretations, most scholars agree that the gradual implementation of "*gaitu guiliu*" significantly reduced the autonomy of hereditary chieftains and marked an important shift toward direct imperial governance in southwestern China (Gong, 2012). The institutional structure and historical evolution of the Tusi system therefore provide an important historical context for understanding how imperial governance operated in multiethnic frontier regions of southwestern China.

2.2 The Targeted Poverty Alleviation

Launched in 2015, the Targeted Poverty Alleviation (TPA) campaign represents a paradigm shift in China's rural governance, moving away from traditional broad-brush redistribution toward a precision-oriented framework. At its core, the TPA emphasizes "seeking truth from facts" and “tailoring interventions to local conditions,” necessitating a scientific procedure to identify, assist, and manage impoverished households based on a granular diagnosis of poverty causes.

The policy architecture is built upon the mandate of “Two Assurances and Three Guarantees” (ensuring adequate food and clothing alongside access to compulsory education, basic medical services, and housing safety). To operationalize this, the program systematically addresses four fundamental questions: “who should be assisted,” “who is responsible for assistance,” “how to assist,” and “how to exit.” This is further supported by the “Five Batches” strategy, which delineates diversified exit pathways including industrial development, involuntary resettlement, ecological compensation, education, and social safety nets. Central to this mechanism is the “Six Precisions” protocol, which functions as a rigorous institutional constraint to ensure that scarce resources are accurately matched with the idiosyncratic needs of the poor. This framework encompasses: precision in identifying poverty targets, precision in tailoring measures to households, precision in project planning, precision in capital allocation, precision in stationing village-level officials (First Secretaries), and precision in evaluating poverty reduction outcomes. Beyond material transfers, the TPA places significant weight on the synergy between “bolstering aspirations” and “enhancing intelligence”. By integrating psychological mobilization with vocational training, this “dual-support” model aims to dismantle the behavioral and cognitive constraints that often trap individuals in persistent poverty (Duflo, 2012), thereby transforming passive relief into endogenous development.

As a primary battlefield of this campaign, Sichuan Province offers a representative setting due to its high concentration of impoverished counties, vast geographical dispersion, and deep-seated deprivation. Among its 88 designated poverty-stricken counties, many are situated in remote, topographically fragmented ethnic regions. In response to such complex geographical and institutional legacies, Sichuan has implemented high-intensity administrative mobilizations that exceed standard policy execution. The successful decertification of all poverty-stricken counties in Sichuan by 2020 provides a unique quasi-natural experiment. This allows us to investigate how such intensive modern interventions—by improving aspiration, skills and Intellectual Empowerment—can effectively counteract the negative persistence of historical institutions (Dell, 2010) and ultimately catalyze firm entry and the revitalization of market dynamism.

2.3 Theoretical framework

State capacity theory is an important theory in social science research. As a classic political theory, it has been widely applied to explain economic and social phenomena. It refers to the ability of the government to effectively implement policies, maintain social order, provide public services, and promote economic development (Skocpol, 1985; Migdal, 1988; Englebert, 2000; Robinson, 2008). State capacity is particularly important in the context of multi-ethnic and decentralized governance, as it directly relates to the coordination and integration between central and local governments, the diffusion of legal systems, and the effective allocation of resources (Yang, 2003).

In the 1980s, state capacity theory gained global attention through Skocpol's research. State capacity is generally defined as the ability of a state to achieve its goals through policy execution and governance actions (Skocpol, 1985), and it specifically includes the state's extractive capacity, coercive capacity, and administrative capacity, such as public service delivery and information collection capacity. The impact of the Tusi system on local governance is reflected through several dimensions. First, extractive capacity: local Tusi lords had significant control over regional resources, but due to the limited actual control by the central government in these areas, local governments were weak in executing and implementing economic policies. This situation led to inefficient resource allocation in the Tusi system regions, which in turn affected the state's governance capacity in those regions. Second, coercive capacity: the Tusi system granted local lords certain military and social control abilities. However, the power of these local leaders often conflicted with the central government's policy objectives, causing instability in local governance. Third, administrative capacity: under the Tusi system, the relative independence of local rulers made it difficult for the central government to effectively implement policies, leading to weak local administrative capacity. This long-standing governance deficiency caused the Tusi system regions to lag behind other regions in the modernization process, with lower state capacity.

On one hand, state capacity can be used as a dependent variable in research. We can analyze how state capacity in areas governed by the Tusi system, such as the Sichuan region, was shaped. State capacity does not naturally form but is gradually shaped by specific historical contexts and social systems. It is the result of political (Acemoglu & Robinson, 2005), war (Tilly, 1992), and historical (Chang, 2019) factors. The Tusi system's feature of local autonomy, particularly the continuation of ethnic autonomy and local rules, limited the downward transmission of state governance capacity and the diffusion of legal systems. Local Tusi lords enjoyed considerable autonomy, creating rules and resource allocation patterns that were in stark contrast to central government policies. This local governance model often resulted in insufficient state governance capacity, with resource distribution, market rules, and administrative efficiency all constrained by the historical legacy of the Tusi system. The legacy of this system is reflected in how local governments support firm entry, enforce the rule of law, and open up markets. Furthermore, the Tusi system historically reinforced local rules and social traditions, making local governance dependent on Tusi leaders rather than the central government. The position of the Tusi leaders relied on their cooperation with the central government, but this cooperation often hinged on local autonomy and flexible handling of affairs, which weakened the central government's control and enforcement capabilities. The Tusi system also allowed long-standing political and economic autonomy at the local level, meaning that local governments often relied on the personal capabilities of Tusi leaders rather than the institutional capacity of government agencies. The long existence of this local political structure led to lower state capacity in Tusi system areas compared to other regions directly governed by the central government. Therefore, the Tusi system weakened the central government's control over local governance, especially in policy execution and the diffusion of legal systems. Through local autonomy, Tusi system regions developed a governance model that was asymmetrical to central policies, and the Tusi system, by granting local lords autonomy, obstructed the centralization process. This historical legacy led to weaker state capacity in Tusi system regions.

On the other hand, state capacity is not only a dependent variable but also a key explanatory variable for economic development (Dincecco & Katz, 2016). The strength or weakness of state capacity directly impacts economic development levels and market stability. Strong state capacity ensures effective resource allocation, stable social order, a sound legal system, and effective public services (Weiss, 1998), all of which contribute to a favorable environment for business development. However, regions with weak state capacity typically face market entry barriers, inefficient resource allocation, and poor legal systems, leading to slower economic development and suppressed firm entry. The governance capacity in Tusi system legacy regions is weaker, leading to higher market entry thresholds and greater entry costs for firms. Enterprises often face an environment with weak policy execution, numerous local rule barriers, and increased market uncertainty, which lowers investment attractiveness (Rodrik, 1991). Therefore, based on the above discussion, the following hypothesis is proposed:

H1: The Tusi system suppressed contemporary firm entry.

Traditional Institutional Economics posits that economic behavior and development are profoundly shaped by the historical evolution of institutions, social customs, and culture. Building upon this, New Institutional Economics categorizes institutions into formal and informal types, scrutinizing their respective impacts on resource allocation, market boundaries, and vertical integration (Rutherford, 1996). A cornerstone of this framework is the Path Dependence Theory proposed by Douglas North (1990), which asserts that historical decisions and events exert a persistent "lock-in" effect on current and future trajectories. Once a specific path is chosen during technological or institutional evolution, the force of inertia ensures its continual reinforcement, thereby underscoring the "primacy of history."

The Tusi system functions as a quintessence of extractive political institutions (Acemoglu & Robinson, 2012), characterized by the concentration of power within a hereditary elite and a profound absence of checks, balances, or the rule of law. Such political structures inevitably give rise to extractive economic institutions, manifested in the collapse of legal order, precarious property rights, and high entry barriers that stifle market mechanisms and foster an unlevel playing field.

Specifically, this system lacks an effective incentive structure, leading to incentive misalignment where individuals are discouraged from pursuing self-interest maximization, while pervasive transaction costs undermine overall economic efficiency. In the context of this study, although the Tusi system originated as a formal administrative arrangement, its legacy persists through informal institutions—such as local power factions and ancestral clan covenants—which continue to dominate contemporary regional financial environments (Guiso et al., 2016). The inherent geopolitical isolation and semi-closed social structures of these regions, coupled with the absence of property rights protection, hindered the early maturation of modern contracting spirits and credit systems (La Porta et al., 1997). Furthermore, the social networks and entrenched power hierarchies formed under this regime represent substantial sunk investments, making the opportunity cost of institutional transition prohibitively high (North, 1990). This institutional persistence places these regions at a distinct disadvantage in modern financial transformation, ultimately manifesting as lower coverage of financial institutions and systemic inefficiency in credit allocation.

Furthermore, intertemporal resource allocation, the essence of modern finance (Campbell & Viceira, 2002), necessitates a robust legal environment and transparent information dissemination. However, the historical dominance of clans and local power in Tusi regions has engendered significant information asymmetry and elevated transaction costs, thereby deteriorating the business environment (Lambert et al., 2012). The entrenched reliance of local actors on "rule-of-man" over "rule-of-law" has been transmitted across generations, resulting in a deficit of modern property rights protection and equitable legal frameworks. This misalignment between historical legal traditions and modern financial requirements (Beck et al., 2003) ultimately stifles firm entry.

Consistent with path dependence, although the Tusi as a political entity was superseded by "*Gaituguiliu*" and modern governance, its centuries-old social norms, clan identity, and contractual habits persist as a form of "cultural inertia" or "social gene." In areas where modern legal reach remains tenuous, these informal rules continue to dictate commercial decisions and credit conditions, suppressing the deep development of financial markets. Modern finance—encompassing both traditional banking and emerging FinTech—is crucial for firm entry due to its control over capital resources. In Tusi-affected regions, the scarcity of modern financial supply compels potential entrants to lower their entry expectations and motivations due to financing constraints, lack of seed capital, and insufficient risk collateral (Aghion et al., 2007), thereby reducing the overall volume of regional firm entry.

Based on the above analysis, this paper proposes the following hypothesis:

H2: The Tusi system inhibits the entry of new firms by suppressing the development of modern finance.

According to Human Capital Theory (Schultz, 1961), a high-quality labor force constitutes the bedrock for firm entry and long-term survival. However, the enduring legacy of the Tusi system may have stifled modern human capital accumulation through two distinct channels.

Primarily, based on the logic of path dependence, the historical rigidity of social strata and the lack of upward mobility in Tusi regions led to chronic underinvestment in educational infrastructure. This semi-enclosed social structure effectively insulated these areas from the penetration of modern educational systems, resulting in a significantly lower regional knowledge stock. Moreover, as a quintessence of traditional Chinese culture, Confucianism historically played a pivotal role in the transmission of education through the imperial examination system and state-sponsored academies. Yet, due to long-standing fragmentation and regional autonomy, Tusi regions were seldom reached by central Confucian orthodoxy and even maintained a wary stance toward its expansion (Feng et al., 2019). Consequently, the absence of this critical ideological and cultural support rendered human capital accumulation relatively stagnant. In parallel, Feng et al. (2019) argue that the Tusi system undermined the capacity-building of the central government and hindered the downward mobilization of elites. Given that the competence of political leadership directly dictates regional developmental prospects and resource allocation (Migdal, 1988), this institutional friction further constrained the provision of educational resources.

Secondly, following the talent allocation framework of Murphy et al. (1991), the remnants of clan-based power and localized networks in Tusi regions likely precipitated a misallocation of human capital. When social capital is predicated on interpersonal ties rather than professional competencies, social elites are incentivized to exhaust their talents in non-productive rent-seeking activities—leveraging entrenched power—rather than venturing into high-risk modern entrepreneurial activities.

Modern enterprises rely on skilled labor and management expertise. Scarce and distorted regional human capital drives up search and training costs, raising operational risks and barriers to entry. As shown in Hopenhayn (1993), frictions that increase firm-level adjustment costs or operational risk can significantly deter firm entry and reduce aggregate employment.

Based on these arguments, we propose:

H3: The Tusi system inhibits firm entry by suppressing the development of human capital.

In the framework of New Economic Geography, infrastructure serves as a quintessential physical substrate for generating agglomeration effects, minimizing transaction costs, and facilitating factor mobility (Krugman, 1991). As a classic public good, the provision efficiency of infrastructure is profoundly dictated by the prevailing institutional arrangements (Samuelson, 1954). Consequently, the inefficient institutional design of the Tusi system may have stifled regional infrastructural development through two distinct dimensions.

First, regarding the idiosyncratic characteristics of the Tusi system, the territorial autonomy exercised by various Tusi jurisdictions historically engendered significant border effects and geopolitical barriers. Consistent with the discourse on public goods provision by Alesina et al. (1999), historical political fragmentation often precipitates a "fragmentation of public decision-making," thereby suppressing the overall supply of public goods. This fractured governance has led to a legacy of disintegrated regional transport and communication networks, leaving these areas underdeveloped in the modern era.

Second, from the perspective of resource allocation capacity, infrastructure—as a quasi-public good—requires sophisticated coordination between the state and the market. However, in regions overshadowed by the Tusi legacy, the allocative efficiency of local authorities is often compromised. These entities are prone to diverting resources toward non-productive, rent-seeking activities aimed at maintaining political power, rather than investing in public infrastructure with strong positive externalities (Aghion & Durlauf, 2013). Such a distortion in fiscal expenditure structures severely constrains the completion of the auxiliary infrastructure essential for modern enterprises.

Ultimately, the lag in infrastructure development directly escalates operational costs for firms, notably in terms of logistics and communication. When the physical environment of a region fails to sustain the supply chain requirements of modern industries, potential entrants confront formidable physical barriers. This lack of spatial connectivity significantly dampens the motivation for market entry (Krugman, 1991).

Based on the aforementioned analysis, this paper proposes the following hypothesis:

H4: The Tusi system hinders firm entry by suppressing the development and connectivity of regional infrastructure

The conceptual framework is presented in Appendix Fig. A.1.

3. Data, variables and empirical strategy

3.1 Sample acquisition and the source of data

To empirically examine the impact of the Tusi system on firm entry, this study constructs a comprehensive panel dataset covering Chinese 183 counties from 2011 to 2023. The primary data sources are categorized as follows: (1) Chinese Research Data Services (CNRDS): This platform provides granular data on listed companies' business operations, banking and financial indicators, and various socio-economic metrics; (2) Industrial and Commercial Registration Database (ICRD): This database comprises comprehensive registration records of Chinese enterprises since 1949. Key variables utilized in this study—including registration dates, ownership types, industry codes, and equity structures—serve to identify firm entry events and calculate the aggregate number of enterprises at the district and county levels; (3) Sichuan Statistical Yearbook: This repository offers multi-dimensional indicators encompassing natural resources, economic performance, social development, and ecological conditions within the region; （4）National Bureau of Statistics of China（NBS）:This department is responsible for overseeing national statistics and national economic accounting, executing national statistical survey plans, and providing population census data. (5) Draft History of Qing (Qing Shi Gao) and General Gazetteer of Sichuan (Sichuan Tongzhi): To capture historical institutional variations, we consult the Draft History of Qing, which provides a definitive record of major historical events and institutional frameworks across the 296-year span of the Qing Dynasty. This is supplemented by the General Gazetteer of Sichuan, a provincial-level historical record from the Qing era that chronicles geographic evolution, fiscal and taxation systems, and frontier military administration, thereby providing a rigorous historical basis for our identification strategy;(6) China Poverty Alleviation database (CPAD): A comprehensive repository that encapsulates historical archives, policy trajectories, and socio-economic indicators of impoverished regions. By integrating expert insights with empirical statistics, the CPAD provides a robust theoretical and data-driven foundation for this study, ensuring the academic rigor and reliability of our analysis on poverty reduction efforts. (7) National Platform for Common Geospatial Information Services (NPCGIS): This platform integrates national fundamental geographic resources and thematic natural resource data. Building upon this platform, we utilized ArcGIS tools for spatial data extraction and processing to obtain precise county-level geographic attributes.

3.2 Variable definitions

3.2.1 Dependent variable: lnfirm

Drawing from the Chinese Industrial and Commercial Registration Database, this paper constructs a proxy for firm entry by taking the logarithm of new firm registrations across various counties (2011–2023). Consistent with the practical framework of Cui and Li (2023), the annual volume of newly registered enterprises serves as a surrogate for entry intensity. We identify the entry timing through a synchronization of commencement and establishment dates, systematically filtering out entries with chronological inconsistencies or missing date parameters. Geographical attribution is conducted via registration authority records; in cases where such data are absent, we decode the six-digit administrative division codes from the business license numbers according to contemporary administrative standards. Any records lacking both administrative and licensing identifiers are treated as noise and removed from the dataset. To address potential heteroscedasticity, the logarithmic value of newly established firms per city-year is adopted as the primary dependent variable.

3.2.2 Independent variable: tusi

Following the empirical framework of Feng et al. (2019), the core explanatory variable in this study is a dummy variable, tusi, representing whether a county was historically under the jurisdiction of the Tusi system (1 if yes, 0 otherwise). This indicator was manually compiled and cross-referenced based on historical records from the Draft History of Qing (Qing Shi Gao) and the General Gazetteer of Sichuan (Sichuan Tongzhi).

3.2.3 Control variables

Following Feng et al. (2019) and Li and Lin (2016), who examine the persistent economic impacts of the Tusi system, this study incorporates a suite of control variables categorized into historical, environmental, and contemporary economic factors to account for potential determinants of firm entry. Regarding historical background, we include chama, a dummy variable indicating whether a county was situated along the Ancient Tea Horse Road[3], to proxy for historical commercial activity. Additionally, lnheat captures the Caloric Suitability Index circa 1500 CE, reflecting the development of ancient smallholder economies (Galor & Özak, 2016); For environmental controls, we employ temperature dummy and precipitation dummy to identify regions with optimal climates for growth (annual temperatures of 10-16°C and precipitation of 800-1500mm), alongside ecoquality as a comprehensive measure of ecological; Lastly, to reflect contemporary economic conditions, we employ the annual secondary industry output (industry), government fiscal revenue (revenue), and government fiscal expenditure (expenditure), each calculated as a percentage of the local GDP (multiplied by 100). Furthermore, urbanization rates (urban) and nighttime light intensity (lnnl) are added as additional proxies for regional development. All data are sourced from the Sichuan Statistical Yearbook and GIS databases.

Appendix Table A.1 summarizes the variable definitions.

3.3 Descriptive statistics

Table 1 reports the descriptive statistics. The wide range of lnfirm reflects significant disparities in regional economic vitality and firm entry across counties. Similarly, the substantial variation in secondary industry output and fiscal expenditure underscores the heterogeneous economic structures and fiscal capacities within the sample. These pronounced cross-sectional variations provide a robust basis for our empirical identification.

[3] The Ancient Tea Horse Road refers to a network of commercial trade routes established since the Tang Dynasty to meet local demands in Southwest and Northwest China. Centered on the exchange of tea and horses, these routes relied on caravans as the primary mode of transportation and served as a vital corridor for economic and cultural exchange among ethnic groups in Southwestern China.

Table 1. Descriptive statistics.

Variables	Variable name	N	Mean	SD	Median	Min	Max
Panel A: Key variables							
Dependent variables	firm	2379	3606.549	5433.088	2290.000	7.000	63529.000
	lnfirm	2379	7.601	1.119	7.737	2.079	11.059
	firm density	2379	1.047	1.083	0.786	0.000	6.691
	ln private value	2379	13.296	1.335	13.585	9.400	16.822
Independent variable	tusi	2379	0.246	0.431	0.000	0.000	1.000
Instrumental variable	wei_distance_altitude	2379	4.600	6.474	0.902	0.001	24.930
Panel B: Mechanism variables							
	financial_outlets_density	2379	0.097	0.247	0.034	0.000	2.139
	state_owned_bank_density	2379	0.054	0.170	0.014	0.000	1.432
	rural_credit_cooperative_density	2379	0.033	0.051	0.018	0.000	0.462
	primary_students_density	2379	2.652	1.339	2.729	0.257	7.629
	middle_students_density	2379	2.383	1.326	2.449	0.034	6.977
	college_students_density	2379	2.860	1.743	2.669	0.143	9.237
	illiterate	2379	8.746	9.224	5.200	1.110	40.090
	ln_hospital	2379	1.433	0.489	1.386	0.000	3.178
	ln_highway	2379	0.742	0.520	0.680	0.049	3.946
	ln_railway	2379	0.067	0.154	0.022	0.000	1.307
	ln_station	2379	0.015	0.074	0.001	0.000	0.638
	ln_comm	2379	4.923	1.785	5.124	0.000	9.237
Panel C: Other variables							
Control variables	chama	2379	0.148	0.355	0.000	0.000	1.000
	lnheat	2379	8.160	0.130	8.150	7.510	8.408
	temperature	2379	0.251	0.433	0.000	0.000	1.000
	precipitation	2379	0.779	0.415	1.000	0.000	1.000
	ecoquality	2379	0.600	0.112	0.592	0.247	0.919
	industry	2379	42.750	16.537	43.339	4.129	85.677
	revenue	2379	5.874	13.353	4.854	0.225	584.020
	expenditure	2379	35.790	41.050	19.406	1.688	394.068
	urban	2379	42.480	19.733	39.920	4.600	100.000
	lnnl	2379	0.434	0.693	0.135	0.000	3.812

3.4 Model setting

To examine the impact of the Tusi system on contemporary firm entry, we specify the following baseline econometric model:

$$lnfirm_{it} = \alpha + \beta tusi_i + \gamma X_{it} + \lambda_t + \epsilon_{it} \quad (1)$$

Here i stands for county i in Sichuan province. t means year t between year 2011 and year 2023. The outcome of interest $lnfirm_{it}$ represents the logged number of newly established enterprises in county i in year t. $tusi_{it}$ is the independent variable which means county i whether was implemented Tusi system before. X_{it} represents a series of control variables which may impact the long-term entrance of firms in county i in year t. We include year fixed effects λ_t to absorb time-varying factors that affect all observations equally within a given year. α and ϵ_{it} stand for the constant and standard errors respectively. To enhance the reliability of our findings, the standard errors are clustered at the county level.

4. Empirical results and analysis

4.1 Baseline regression results

The baseline regression reveals a significant negative impact of the Tusi system on regional firm entry. Table 2 presents the baseline regression results. In the Column (1), the tusi coefficient stands at -1.299, indicating that regions historically under the Tusi system exhibit substantially lower firm entry rates than their counterparts. This negative effect remains robust and statistically significant at -0.240 in Column (6) after sequentially incorporating a comprehensive set of controls, implying that compared to regions with conventional institutional histories, those with a legacy of the Tusi system experience a 24% lower level of contemporary firm entry on average. The stability of this coefficient suggests that the inhibitory effect of the Tusi system is not merely driven by observable geographic or economic covariates but possesses distinct independence and persistence. Economically, these results underscore the enduring legacy of historical institutions; the long-standing political and economic insularity associated with the Tusi system may have hindered the maturation of market mechanisms, thereby creating barriers to contemporary firm entry.

The variable representing the Ancient Tea Horse Road (chama) is statistically significant at the 1% level in Column (2), but becomes insignificant in Columns (5) and (6) after sequentially adding industry, fiscal, and regional controls. This pattern suggests that the historical trade route's direct effect on new firm entry is largely absorbed by contemporary structural factors—such as industrial structure, fiscal capacity, and urbanization levels. While the route once underpinned regional economic integration, these legacy effects appear to be mediated by modern institutional and economic conditions, rather than exerting an independent, persistent influence on firm entry.

Environmental controls—including precipitation (precipitation), temperature (temperature), and ecological quality (ecoquality)—exhibit heterogeneous effects. Precipitation enters positively and significantly at the 1% level across specifications, indicating that favorable water availability supports primary production and resource-based activities, thereby facilitating new firm entry. Temperature, however, is statistically insignificant in all models, suggesting that temperature fluctuations do not exert a robust, independent effect on firm entry. In contrast, ecological quality (ecoquality) carries a significantly negative coefficient (at the 1% level in Columns 3 and 6). This counterintuitive finding likely reflects stringent environmental regulatory constraints: ecologically vulnerable or high-quality areas often face stricter resource-use limits and development restrictions, which compress the operational scope for new enterprises, outweighing any potential benefits of natural amenities for firm entry.

Regarding contemporary economic indicators, the coefficient for urbanization (urban) is consistently positive, suggesting that rapid urbanization enhances regional attractiveness by expanding market scale, improving infrastructure, and increasing labor mobility. Conversely, the secondary industry output (industry) yields a negative coefficient significant at the 1% level. This implies that industrial expansion may raise entry barriers, potentially due to intensified market competition, higher sunk costs, or the crowding out of infrastructure and resources by incumbents. The coefficient for fiscal revenue (revenue) is statistically insignificant at -0.002, whereas fiscal expenditure (expenditure) negatively impacts firm entry at the 1% significance level. This suggests that excessive government intervention or misallocated fiscal resources may stifle entrepreneurial dynamics. Nighttime lights (lnnl), a proxy for economic intensity, show a significantly positive correlation with firm entry, reflecting that robust market demand and superior infrastructure in economically active areas provide fertile ground for business development.

Table 2. Baseline results.

	(1)	(2)	(3)	(4)	(5)	(6)
	lnfirm	lnfirm	lnfirm	lnfirm	lnfirm	lnfirm
tusi	-1.299***	-1.014***	-0.675***	-0.383***	-0.367***	-0.240**
	(0.139)	(0.155)	(0.144)	(0.104)	(o.111)	(0.101)
chama		-0.851***			-0.129	0.002
		(0.228)			(0.117)	(0.112)
lnheat		-0.029			0.154	-0.243
		(0.501)			(0.308)	(0.344)
temperature			0.200			0.135
			(0.153)			(0.085)
precipitation			0.968***			0.494***
			(0.148)			(0.087)
ecoquality			-3.677***			-1.534***
			(0.632)			(0.525)
industry				-0.010***	-0.010***	-0.007**
				(0.003)	(0.003)	(0.003)
revenue				-0.002	-0.002	-0.002
				(0.003)	(0.003)	(0.003)
expenditure				-0.014***	-0.014***	-0.012***
				(0.002)	(0.002)	(0.002)
urban				0.005	0.006	0.008**
				(0.004)	(0.004)	(0.004)
lnnl				0.524***	0.521***	0.324***
				(0.081)	(0.082)	(0.096)
Time FE	YES	YES	YES	YES	YES	YES
Obs.	2,379	2,379	2,379	2,379	2,379	2,379
R-squared	0.325	0.386	0.541	0.720	0.722	0.753

Notes: Standard errors, clustered at the county level, are reported in parentheses. *Significant at 10%, **Significant at 5%, ***Significant at 1%. The same applies to the table below.

4.2 Robustness tests

4.2.1 Endogenous test

In the baseline regression, we included various controls for historical endowments, natural environments, and modern economic factors to mitigate omitted variable bias. However, a key geographic factor—altitude—remained unaddressed. Sichuan Province features highly diverse topography, including plateaus, mountains, hills, and plains, with a significant elevation gradient from west to east. Such heterogeneous geographic conditions likely exert a direct influence on the location and entry decisions of contemporary firms. To further address the potential endogeneity of our core explanatory variable, we combine historical institutional layouts with natural geographic endowments, using the interaction between the logarithm of the linear distance from a county's geometric center to the nearest Ming Dynasty *Weisuo*[4] (military garrison) and the county's mean altitude as an instrumental variable (IV) for the Tusi system.

The validity of this instrument is based on two arguments. First, regarding relevance, natural geographic conditions directly determined the frontier governance costs for ancient central governments. High altitude significantly increased the difficulty of transportation and logistics, while greater distance from a *Weisuo* garrison signaled weak administrative and military control. Regions characterized by both high altitude and isolation from military outposts became governance blind spots where central authority struggled to penetrate. Given the prohibitive costs of direct rule, the government favored the Tusi system in these rugged and remote areas, relying on local tribal elites for indirect administration. This interaction term precisely captures the dual impact of geographic barriers and historical military layouts, effectively mapping the influence of geographic constraints on the formation and persistence of the Tusi system and reinforcing the effects of institutional path dependency. Thus, it maintains a strong theoretical and empirical correlation with the endogenous variable.

Second, regarding the exclusion restriction, *Weisuo* were purely military defense facilities established over 600 years ago based on the military logic of Ming-era frontier defense; their locations bear no direct relation to the operational or locational choices of modern micro-enterprises. Similarly, while mean altitude is an innate natural endowment that may indirectly affect production, its impact—once temperature, modern economic variables, and time fixed effects are controlled for—is unlikely to influence modern firm entry through any channel other than the historical Tusi system. Therefore, the interaction term effectively isolates the direct interference of natural geography and historical military positioning, ensuring the instrument affects contemporary firm entry only through the core path of the historical Tusi system, thereby satisfying the exclusion restriction.

Table 3 presents the two-stage least squares (2SLS) estimation results. The first-stage regression (Column 1) yields a significantly positive coefficient for the interaction instrument, confirming the relevance assumption. In the second-stage regression (Column 2), the estimated coefficient of Tusi system remains negative and statistically significant at the 1% level, which further corroborates the robustness of baseline findings. Meanwhile, the Kleibergen-Paap rk LM

[4] The *Weisuo* (Garrison) system was a pivotal military and economic institution in the Ming and Qing dynasties. Established by Emperor Zhu Yuanzhang based on the *Fubing* model, its core logic was "integrating soldiers with farmers" to maintain a self-sufficient defense force through state-sponsored farming (*Tuntian*). Each Guard (*Wei*) comprised 5,600 hereditary soldiers, strategically stationed across the empire. While the system declined in the late Ming due to land encroachment and desertion, the Qing Dynasty retained the framework, shifting its function from military defense to administrative land taxation and civil management.

statistic rejects the under-identification hypothesis, and the Cragg-Donald Wald F statistic far exceeds the critical value for weak instrument test, verifying that our instrument is valid and strong.

Table 3. Endogenous test.

	(1)	(2)
	First stage	Second stage
	tusi	lnfirm
wei_distance_altitude	0.058^{***}	
	(0.009)	
tusi		-1.507^{***}
		(0.283)
Controls	YES	YES
Time FE	YES	YES
Kleibergen-Paap rk LM statistic	27.653^{***}	
Kleibergen-Paap rk Wald F statistic	44.063	
（Stock-Yogo 10% critical value）	(16.38)	
Obs.	2,379	
R-squared	0.564	

4.2.2 Accounting for geographic factors

Since the Tusi regions are predominantly located in the high-altitude areas of western Sichuan, we employ a residualization approach to disentangle the institutional effects of the Tusi system from its high-altitude geographical features, thereby mitigating potential identification bias arising from their high correlation.

Specifically, we first regress altitude on the tusi and all other control variables to extract an altitude residual that is orthogonal to the institutional setting. This residual, representing the “pure” geographical endowment, is then included as a control variable in our primary specification. The results in Table 4 indicate that the coefficient for the Tusi system remains negative and statistically significant at the 1% level. Similarly, the altitude residual is also significantly negative at the 1% level. These findings suggest that the long-term inhibitory effect of the Tusi system is robust even after accounting for geographical constraints, while confirming that high-altitude conditions themselves serve as a substantial barrier to firm entry.

Table 4. Accounting for geographic factors.

	(1) altitude	(2) lnfirm
tusi	2.118***	-0.732***
	(0.156)	(0.107)
altitude_resid		-0.486***
		(0.053)
Historical control variables	YES	YES
Environmental control variables	YES	YES
Contemporary control variables	YES	YES
Time FE	YES	YES
Obs.	2,379	2,379
R-squared	0.543	0.801

4.2.3 Accounting for cultural influences

Given that Sichuan is a multi-ethnic region with diverse cultural and religious traditions, we further control for cultural factors to exclude their potential influence on firm entry. Following Kung and Ma (2014), we include three proxy variables: a dummy variable for the presence of Confucian Temples, and the densities of Buddhist and Taoist cultural sites, calculated as log (count/area + 1). The results in Table 5 show that the coefficient of the Tusi system remains negatively significant even after accounting for these cultural and religious dimensions, thereby reinforcing the robustness of our baseline findings.

Table 5. Accounting for cultural influences.

	(1) lnfirm	(2) lnfirm	(3) lnfirm
tusi	-0.221**	-0.237**	-0.238**
	(0.100)	(0.102)	(0.101)
Confucianism	0.190***		
	(0.064)		
Buddhism_desity		0.761	
		(3.322)	
Taoism_density			7.753
			(14.386)
Historical control variables	YES	YES	YES
Environmental control variables	YES	YES	YES
Contemporary control variables	YES	YES	YES
Time FE	YES	YES	YES
Obs.	2,379	2,379	2,379
R-squared	0.756	0.753	0.753

4.2.4 Considering the influence of ethnic minorities

The high concentration of ethnic minorities in Sichuan Province presents a potential confounding factor that may influence our empirical results. Given that our baseline model does not explicitly account for ethnic composition, we incorporate the minority population share—sourced from the 2020 Seventh National Population Census—as a formal control variable. Since firm entry is a contemporary phenomenon and the tusi variable identifies a persistent institutional legacy, utilizing modern demographic proportions as a proxy is theoretically consistent with the literature on historical persistence.

The results are reported in Column (1) of Table 6. Upon controlling for the minority share, the coefficient for tusi decreases in magnitude and loses statistical significance. However, this is likely an artifact of the tusi system's inherent design, which was specifically engineered for the effective governance of ethnic minorities in frontier regions. Consequently, historical tusi jurisdictions in Southwest China, including Sichuan, exhibit a high degree of spatial overlap with modern ethnic enclaves. This suggests that the minority proportion is not an exogenous factor independent of the tusi system; rather, it is endogenous to the institutional treatment. Including it as a control variable may therefore introduce over-controlling bias, leading to an underestimation of the total effect of the institutional legacy.

To address this, we conduct a robustness check by substituting the explanatory variable with the minority share (minority) as an alternative proxy for the tusi legacy. This approach introduces an intensity measure that more accurately captures the depth and penetration of historical institutional influence. The results are reported in Column (2) of Table 6. The results remain robust after this substitution, further validating the reliability of our core findings.

4.2.5 Replace the dependent variable

To ensure the reliability of our findings, we conduct robustness checks by employing alternative measures of firm entry. In Column (3) of Table 6, we replace the dependent variable with "firm density"—defined as the number of firms per unit of land area—to account for geographic scale effects. Furthermore, considering that the vast majority of new entries in Sichuan (2011–2023) are private enterprises, we use the logarithm of the added value of the private economy as a surrogate dependent variable in Column (4). The results show that the coefficients for Tusi remain negative and statistically significant at the 1% level across both specifications. The consistency in sign and significance with the baseline results reinforces the conclusion that the historical Tusi system exerts a robust inhibitory effect on contemporary entrepreneurial activity.

4.2.6 Replace the estimation model

Given that the raw data consists of count variables, we employ appropriate count models for further validation. Descriptive statistics indicate that the variance of firm registrations significantly exceeds the mean, suggesting potential over-dispersion. To address this, we adopt a Negative Binomial model instead of a Poisson specification to achieve better fit and accuracy. The results, presented in Column (5) of Table 6, show that the coefficient for Tusi remains highly significant and consistent with the baseline estimates. This reaffirms that the Tusi system consistently inhibits firm entry, further strengthening the robustness of our primary findings.

4.2.7 Placebo test

To ensure that our baseline results are not driven by spurious correlations or accidental matching, we conduct a permutation-based placebo test for the tusi variable. Specifically, while holding the dependent variable and other controls constant, we randomly reassign the tusi status across counties. We repeat this procedure 500 times, re-estimating the baseline regression in each iteration to obtain a distribution of "pseudo-tusi" coefficients. Figure 3 presents the kernel density distribution of these 500 estimated coefficients (indicated by the pink dots and solid curve). The distribution is centered around zero, whereas our actual baseline estimate (the vertical line at −0.240) lies at the extreme left tail, far removed from the null distribution. In other words, the probability of obtaining a negative effect of such magnitude through random assignment is negligible. This confirms that the negative association between the tusi system and firm entry is not a product of random noise or specific sample partitions, but rather possesses substantial statistical robustness.

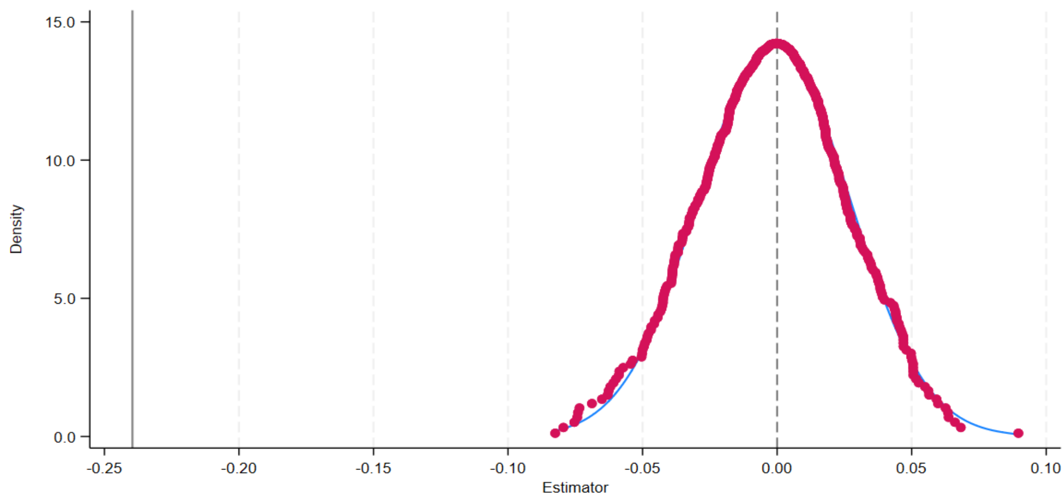

Fig. 3 Placebo test

Notes: This figure plots the kernel density distribution of 500 "pseudo-tusi" coefficients from a permutation-based placebo test. We randomly reassign the tusi status across counties while keeping the dependent variable and all controls unchanged. The vertical line represents our actual baseline estimate (-0.240).

Table 6. Robustness check.

	(1) lnfirm	(2) lnfirm	(3) firm density	(4) ln private value	(5) firm
tusi	0.122		-0.232***	-0.403***	-0.258**
	(0.117)		(0.057)	(0.113)	(0.109)
minority	-0.010***	-0.009***			
	(0.003)	(0.002)			
Controls	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES
Obs.	183	183	2,379	2,379	2,379
R-squared	0.817	0.816	0.793	0.846	

5. Mechanism analysis and heterogeneity analysis

5.1 Mechanism analysis

Building upon the theoretical framework, this section aims to elucidate the internal logic through which the Tusi system influences contemporary firm entry. To test the proposed hypotheses, we employ a mediation effect framework. Specifically, we construct intermediary indicators representing regional financial development, human capital accumulation, and infrastructure quality to scrutinize the transmission channels through which historical institutional legacies shape modern micro-economic behavior.

Broadly speaking, the regional business environment serves as the pivotal determinant of firm entry. From a factor endowment perspective, this study categorizes the aforementioned mechanisms into two dimensions: "Soft Factor Environment," referring to the supply quality of capital and labor represented by modern finance and human capital; and "Hard Entry Barriers," referring to physical constraints such as infrastructure. This classification elucidates how the Tusi system stifles contemporary entrepreneurial intent—firstly by undermining the "intangible" efficiency of factor allocation, and secondly by restricting the "tangible" physical foundations necessary for market activity.

5.1.1 The modern finance mechanism

Despite the progressive efforts of the Yuan, Ming, and Qing dynasties to tighten control, governance in Tusi regions remained markedly weaker compared to areas under direct imperial administration. This governance gap impeded the effective penetration of formal state regulations into the local social fabric. Under conditions of limited external intervention, non-institutionalized traditions were allowed to flourish and persist (Greif, 2006). Within this relatively isolated frontier environment, indigenous norms gradually consolidated, creating a latent misalignment with the social and legal standards required by a modern socialist market economy.

From this perspective, we examine the transmission of non-institutionalized traditions through the lens of finance—a pivotal factor shaping firm entry. While existing literature extensively documents that robust modern financial environments facilitate firm registration by easing credit constraints ((Berger & Black, 2011; Huang et al., 2026), we shift our focus to the idiosyncratic constraints within former Tusi regions. In these areas, social structures are predominantly organized around kinship and lineage, fostering a culture of low generalized trust toward external entities. Consequently, internal financial mechanisms and capital flows remain heavily reliant on parochial kin-based networks, which are typically associated with low levels of generalized trust and underdeveloped formal financial systems (Guiso et al., 2004).

To empirically test this mechanism, we construct three proxy variables for the modern financial environment: financial outlet density, rural credit cooperative (RCC) density, and state-owned bank density—all calculated as the count of branches per unit of land area. Within the regional financial landscape, RCCs and state-owned banks represent the two most predominant types of institutions. Table 7 reports the mechanism testing results. Columns (1) and (2) demonstrate that the Tusi legacy significantly reduces the density of financial outlets, thereby impeding firm entry. These findings provide preliminary evidence for the financial intermediation channel, supporting Hypothesis 2.

However, further structural analysis reveals compelling heterogeneity. While the Tusi legacy shows no significant impact on the density of state-owned commercial banks (Column 3), it exerts a negative effect on rural credit cooperative (RCC) density, significant at the 1% level (Column 4), which subsequently hampers firm entry (Column 5). This divergence can be attributed to two factors. First, as pillars of the national financial system, state-owned banks expand following a “top-down” administrative logic. Their branch placement is dictated by administrative hierarchy and national strategic planning, often incorporating policy-driven mandates. This compulsory institutional arrangement renders them resilient to local informal constraints. Second, unlike large-scale banks, RCCs are deeply embedded in grassroots communities. This logic aligns with Turvey & Kong (2010), who argue that rural credit activities in China are closely intertwined with local social trust and informal lending networks, and that informal institutional factors significantly shape the operational efficiency of formal rural financial institutions such as rural credit cooperatives.

Table 7. Regression result of the modern finance mechanism.

	(1)	(2)	(3)	(4)	(5)
	financial_outlets_density	lnfirm	state_owned_bank_density	rural_credit_cooperative_density	lnfirm
tusi	-0.853***		0.008	-0.010***	
	(0.136)		(0.011)	(0.004)	
financial_outlets_density		0.244***			
		(0.034)			
rural_credit_cooperative_density					2.553***
					(0.838)
Controls	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES
Obs.	2,366	2,366	2,379	2,379	2,379
R-squared	0.748	0.787	0.673	0.561	0.753

5.1.2 The human capital mechanism

Human capital is a cornerstone of firm entry decisions (Schultz, 1961), with its impact manifesting not only in the "quantity" of schooling years but also in the "quality" of talent allocation. Historically, the fragmented governance of Tusi regions led to a significant lag in the absorption of modern educational systems. More critically, the legacy of informal power structures has distorted the incentive compatibility mechanisms for social elites. Instead of productive entrepreneurship, talent is often diverted toward non-productive sectors, such as clan-based interest maintenance or administrative rent-seeking. This structural misallocation of human capital not only lowers the overall skill level but also diminishes the region's attractiveness to modern enterprises (Braunerhjelm & Lappi, 2023）.

To comprehensively examine the mediating role of human capital, we follow Feng et al. (2019) in constructing a multi-dimensional indicator system encompassing both "educational accumulation" and "health capital." Specifically, we use the density (log-transformed) of primary and secondary school students per unit of area to capture the quantitative expansion of basic education. To capture the qualitative dimension, we utilize 2020 Census data to calculate the density of individuals with tertiary education and above, as well as illiteracy rates. Furthermore, the number of hospitals per county is employed as a proxy for health capital to complement the educational metrics.

Table 8 presents the results. Columns (1) – (4) show that the Tusi legacy significantly suppresses the development of basic education, thereby exerting a negative shock on firm entry. Columns (5) – (8) further delineate the "quality" of human capital. The results consistently demonstrate that former Tusi regions face a dual challenge: a scarcity of high-level talent (tertiary education density) and a persistent prevalence of illiteracy. This structural deficit in knowledge accumulation constitutes a formidable barrier to modern enterprise entry.

In Column (9), we employ the number of hospitals as a proxy for health capital. However, the regression reveals that the Tusi legacy has no statistically significant impact on the distribution of medical resources. This non-significant result may stem from two factors. First, unlike primary and secondary schools—which are deeply embedded in local social structures—the allocation of medical resources is heavily dictated by top-down state healthcare planning and administrative mandates. Second, the "quantity" of hospitals may fail to accurately capture the "health efficiency" that truly matters for firm entry. Entrepreneurial decisions prioritize the actual quality of the labor force; a mere count of facilities, if lacking high-level medical personnel or advanced equipment, may not effectively translate into a tangible health capital advantage.

Table 8. Regression result of the human capital mechanism.

	(1) primary_students_density	(2) lnfirm	(3) middle_students_density	(4) lnfirm	(5) college_students_density	(6) lnfirm	(7) illiterate	(8) lnfirm	(9) ln_hospital
tusi	-0.582***		-0.646***		-0.568***		3.025**		-0.076
	(0.138)		(0.124)		(0.128)		(1.217)		(0.074)
primary_students_density		0.299***							
		(0.039)							
middle_students_density				0.354***					
				(0.042)					
college_students_density						0.294***			
						(0.047)			
illiterate								-0.020**	
								(0.008)	
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Obs.	2,379	2,379	2,379	2,379	183	183	183	183	183
R-squared	0.696	0.789	0.713	0.803	0.851	0.838	0.809	0.804	0.406

5.1.3 The infrastructure mechanism

Beyond the aforementioned "soft" factor constraints, infrastructural deficiencies constitute a formidable "hard" barrier to firm entry. As public goods with strong externalities, the supply efficiency of infrastructure relies heavily on the coordination capacity of administrative power. Historically, the "territorial autonomy" characteristic of the Tusi system resulted in significant spatial governance fragmentation. Given that cross-regional projects—such as provincial highways and power grids—require high-level political synergy, former Tusi regions often face a legacy of localized power divisions that hampers the efficient provision of public goods (Alesina et al., 1999). This historically rooted infrastructural gap directly escalates logistics and information costs for enterprises. When a region's physical infrastructure fails to integrate seamlessly with national supply chain networks, the incentive for firm entry is severely stifled.

To test this, we construct proxy variables for both transport and communication infrastructure. For transport infrastructure, we measure density (log-transformed) using three indicators: road mileage, railway mileage and railway station counts, all normalized by county land area. These high-resolution geospatial data were retrieved from OpenStreetMap (OSM). Additionally, we utilize mobile phone subscriber density (log-transformed) as a proxy for communication infrastructure to capture the digital connectivity of the region. Table 9 presents the results, showing how these physical constraints, inherited from historical spatial fragmentation, continue to shape the cost structure for contemporary firm entry.

Table 9 reports the empirical results. Columns (1)–(2) indicate that the Tusi legacy significantly reduces road density, thereby suppressing firm entry. This confirms that historical institutional remnants impede the refinement of local transportation networks, escalating logistics costs for enterprises. Furthermore, Columns (5)–(6) reveal a significant negative shock of the Tusi system on mobile phone subscriber density, suggesting that its inhibitory effects have extended from physical space to the digital realm, where lagging communication infrastructure hampers information flow efficiency.

Conversely, Columns (3)–(4) show that the Tusi legacy has no significant impact on the density of railways and railway stations. This divergence may be attributed to the nature of these facilities as “national arteries.” Their planning and site selection are highly centralized under national or provincial strategic mandates, prioritized for macro-resource allocation rather than local social networks. In contrast, road construction relies more on county-level fiscal capacity and land acquisition coordination. In former Tusi regions, fragmented informal power structures may undermine the incentives and efficiency of local governments in providing public goods. Additionally, the unique topography of Sichuan plays a role: railway layouts in Tusi areas are primarily constrained by extreme gradients and geological conditions. When physical geography acts as a “first-order” determinant, the explanatory power of historical institutions is diluted. Roads, being more adaptable to terrain, thus better reflect the institutional disparities in human-led infrastructure development.

Table 9. Regression result of the infrastructure mechanism.

	(1)	(2)	(3)	(4)	(5)	(6)
	ln_highway	lnfirm	ln_railway	ln_station	ln_comm	lnfirm
tusi	-0.309***		0.006	0.005	-0.751***	
	(0.054)		(0.011)	(0.006)	(0.155)	
ln_highway		0.349***				
		(0.071)				
ln_comm						0.166***
						(0.036)
Controls	YES	YES	YES	YES	YES	YES
Constant	0.228	10.561***	-1.033**	-0.567**	3.975	9.943***
	(1.526)	(2.488)	(0.440)	(0.243)	(4.713)	(2.293)
Time FE	YES	YES	YES	YES	YES	YES
Obs.	2,379	2,379	183	183	2,379	2,379
R-squared	0.438	0.763	0.598	0.593	0.637	0.774

5.2 Heterogeneity analysis

To further investigate whether the inhibitory effect of the Tusi system varies with historical governance intensity, we manually compiled two characteristic indicators—the number of Tusi (Num) and the Tusi rank (Rank)—based on the Draft History of Qing and General Gazetteer of Sichuan. The Rank variable is coded on a scale from 1 (e.g., Tu Zhishi) to 9 (e.g., Xuanweishi), following the Qing dynasty's official bureaucratic hierarchy. Table 10 reports the results of the interaction between the Tusi dummy and these characteristics. The interaction term tusi and num lacks statistical significance, suggesting that the negative impact of the Tusi system is not a function of its geographic "breadth" or density. In contrast, the interaction tusi and rank is significantly negative at the 5% level. This implies that the higher the historical administrative rank, the stronger the persistent inhibitory effect on contemporary firm entry. A plausible explanation is that Tusi rank was historically indexed to population size, resource mobilization capacity, and political authority. Higher-ranking Tusi possessed more sophisticated administrative structures and deeper control over their subjects, allowing informal norms—such as clan power and patron-client dependencies—to become more deeply entrenched. These legacies create a significant departure from modern market-contractual spirits, thereby deterring firm entry.

Detailed estimates are reported in Appendix Table A.2.

6.Extended analysis

6.1 Triple-Differences (DDD) Framework

To examine the offsetting and mitigating effects of Targeted Poverty Alleviation (TPA) policies on the negative impacts of the Tusi system, this paper employs a DDD model for further analysis and discussion. The model constructed in this study is as follows:

$$lnfirm_{it} = \beta_0 + \beta_1(Post_t \times Poor_i \times Tusi_i) + \beta_2(Post_t \times Poor_i)$$
$$+\beta_3(Post_t \times Tusi_i) + \beta_4(Poor_i \times Tusi_i) + \gamma_1 Poor_i + \gamma_2 Tusi_i + \gamma_3 X_{it} + \lambda_t + \epsilon_{it} \quad (2)$$

$$P(Poor_i = 1) = \wedge(\alpha + \beta Tusi_i + \gamma X_i + \epsilon_i) \quad (3)$$

This paper constructs a baseline regression model to investigate the impact of Targeted Poverty Alleviation (TPA) policies on firm entry. The dependent variable, $lnfirm_{it}$, is the natural logarithm of the number of newly registered firms in region i during year t. The core explanatory variable is the triple interaction term $Post_t \times Poor_i \times Tusi_i$, whose coefficient β_1 captures the net effect of TPA on firm entry in poverty-stricken counties historically influenced by the Tusi system.

Specifically, $Post_t$ is the policy dummy variable, taking a value of 1 for the period after the implementation of the TPA policy (2015 and onwards) and 0 otherwise; $Poor_i$ is a regional grouping dummy variable, taking a value of 1 for poverty-stricken counties and 0 for non-poverty counties; $Tusi_i$ is a historical institution dummy variable, taking a value of 1 for regions historically affected by the Tusi system and 0 for unaffected regions. The model controls for the static cross-regional differences in poverty status and the Tusi system through $Poor_i$ and $Tusi_i$. Additionally, time fixed effects λ_t and a series of time-varying control variables X_{it} are included to exclude interference from macro-shocks and time-varying regional characteristics

In our model specification, we focus on the following two points: First, the potential overlap and multicollinearity between policy shocks and grouping variables. After 2015, both poverty and non-poverty counties may have been affected by relevant policies, with only the intensity of the policy differing. On one hand, Targeted Poverty Alleviation is not limited to assistance at the county level but provides targeted support down to the household and individual levels. On the other hand, TPA is a nationwide governance mobilization, although the policy focuses on poverty-stricken counties, the resulting infrastructure construction, financial resource allocation, and administrative evaluation pressures all have region-wide spillover effects. Therefore, by using a triple-differences structure, this paper identifies the net effect of the policy on firm entry for specific groups while controlling for universal macroeconomic shocks.

Second, the time-invariant specification of the grouping variable $Poor_i$. This paper sets the poverty county indicator as time-invariant; even if some counties shook off their poverty status during the sample period, they are still consistently marked as $Poor_i = 1$ in the empirical analysis.

This is because what we identify is the policy intervention effect brought by the status of being designated as a poverty-stricken county. The preferential policies, supporting measures, and their externalities enjoyed by poverty-stricken counties are persistent. Meanwhile, the continuous assistance implemented by the government after poverty exit to prevent a return to poverty can also be seen as an extension of the original policy effect.

Furthermore, this paper fully considers the endogeneity issues caused by sample selection bias. The designation of poverty-stricken counties is not determined randomly. If regions influenced by the *Tusi* system were more likely to be classified as poverty-stricken due to historical characteristics such as geographical isolation and weak infrastructure, an endogenous correlation would exist between $Poor_i$ and $Tusi_i$. To address this, this paper incorporates the Logit estimation from Equation (3) to mitigate the endogeneity issues arising from poverty-stricken county grouping, ensuring the robustness and reliability of the baseline regression results.

6.2 Parallel Trend Test

We implement an event-study specification to assess the validity of the parallel trend assumption underlying the DDD design. The estimates indicate no statistically significant pre-policy differences between treatment and control groups. Detailed results are presented in Appendix Fig. A.2.

6.3 Empirical Results and Analysis

A potential concern is that the increase in firm entry in "Tusi poverty counties" might not be due to the policy offsetting the historical legacy, but simply because these areas started from the lowest baseline and rebounded fastest under infrastructure investment. To rule out this sample selection bias, we examine the link between the historical Tusi system and contemporary poverty designation. Our regression results in Table 10 show that, after controlling for relevant variables, the impact of $Tusi_i$ on $Poor_i$ is not statistically significant. This suggests that while the Tusi system was predatory in the past, whether a county is identified for TPA today depends more on contemporary economic standards. This independence provides a strong quasi-natural experimental environment for our DDD model, ensuring it can effectively separate the interaction between historical legacies and modern interventions.

The baseline DDD results (Column 2) show that the coefficient for the core interaction term ($Post_t \times Poor_i \times Tusi_i$) is significantly positive. This indicates that the TPA policy effectively weakened the lingering negative effects of the Tusi legacy and promoted new firm entry. Furthermore, our analysis confirms that TPA offsets historical burdens by combining "intelligence" with "aspiration": the policy significantly increased middle school student density, enhancing human capital while boosting the number of small businesses and self-employed individuals which result from the strengthened entrepreneurial confidence. Together, these findings demonstrate that

TPA successfully revitalizes the regional economy by addressing both the intellectual and motivational barriers inherited from the Tusi system.

Table 10. Extended analysis.

	(1) poor	(2) lnfirm	(3) Middle school student	(4) Individual firm	(5) Small firm
$Post_t \times Poor_i \times Tusi_i$		0.699***	0.224*	0.673***	0.841***
		(0.159)	(0.120)	(0.182)	(0.220)
$Poor_i \times Tusi_i$		-0.633***	-0.437*	-0.684***	-0.835***
		(0.228)	(0.234)	(0.224)	(0.303)
$Post_t \times Tusi_i$		-0.305***	-0.002	-0.263**	-0.425***
		(0.111)	(0.094)	(0.115)	(0.158)
$Post_t \times Poor_i$		0.031	0.128*	0.031	-0.060
		(0.069)	(0.071)	(0.074)	(0.091)
$Tusi_i$	-0.610	0.057	-0.479***	0.076	0.212
	(0.635)	(0.146)	(0.164)	(0.140)	(0.194)
$Poor_i$		-0.118	-0.462**	-0.120	-0.005
		(0.147)	(0.178)	(0.149)	(0.161)
Controls	YES	YES	YES	YES	YES
Constant	-43.803**	9.907***	-0.782	9.515***	10.065***
	(16.690)	(2.665)	(3.092)	(2.671)	(2.889)
Time FE	NO	YES	YES	YES	YES
Obs.	2,379	2,379	2,379	2,379	2,379
R-squared		0.761	0.729	0.737	0.696

7. Conclusions and discussions

7.1 Research conclusion

This study investigates the historical roots of modern micro-economic behavior by examining the persistent impact of the Tusi system on contemporary firm entry. Sichuan, characterized by the long-term co-existence of Tusi autonomy and centralized imperial rule, provides an ideal "natural laboratory" for this inquiry. Utilizing a matched dataset of 182 counties (2,379 observations), we find that the historical Tusi legacy significantly inhibits modern entrepreneurial activity. Quantitatively, the historical legacy of the Tusi system is associated with a 24% reduction in modern firm entry relative to regions governed by conventional administrative systems. This finding remains robust across various tests, including alternative measures of firm entry, different estimation models, and the exclusion of major exogenous shocks. To address potential endogeneity, we employ the interaction between the log-distance from the county centroid to the nearest Ming Dynasty Wei (military garrison) and local altitude as an instrumental variable, which further confirms the baseline results.

Furthermore, we identify the dual-pathway mechanism driving this inhibitory effect: first, through "soft factor constraints," the Tusi legacy hampers the development of modern financial markets and human capital accumulation; second, through "hard investment barriers," it obstructs infrastructure construction. Finally, heterogeneity analysis reveals that the intensity of this suppression is positively correlated with the historical administrative rank of the Tusi, whereas the sheer number of Tusi does not exert such an effect. These findings underscore the profound path dependence of historical institutions in shaping modern market dynamics. Furthermore, this paper conducts an extended analysis using a DDD model. The findings suggest that Targeted Poverty Alleviation policies effectively offset the negative impact of the Tusi system on the entry of modern enterprises, thereby validating the role of contemporary policy interventions in mitigating the legacy of historical institutions.

7.2 Implications

7.2.1 Theoretical Implications

This study constructs a comprehensive framework linking the historical Tusi system to contemporary firm entry, identifying modern finance, human capital, and infrastructure as pivotal transmission mechanisms. By focusing on the tension between the informal norms generated by the formal Tusi institution and modern market mechanisms, this research underscores the enduring legacy of historical institutions on modern economic activity.

Crucially, by distinguishing the empirical effects on state-owned vs. rural credit banks, and national vs. local infrastructure, this paper proposes a significant theoretical corollary: the impact of historical institutions on the modern economy is not universal but "selective." Grassroots organizations that rely heavily on local social capital and community networks are more susceptible to the negative interference of historical traditions. This provides a novel theoretical entry point for explaining why certain modernization policies "fail" in peripheral regions. Furthermore, departing from prior literature that focuses on the geographic "breadth" of institutional distribution, our heterogeneity analysis demonstrates that the "hierarchical depth" of historical administrative power is the core determinant of institutional persistence. Higher-level governance structures imply stronger resource mobilization and more solidified authority orders, creating a social "closed-loop" that is more resilient to the dilutive effects of modern contractual logic. These findings offer a new dimension and direction for the study of historical institutional persistence.

7.2.2 Practical Implications

Our empirical findings suggest that the legacy of the Tusi system continues to hinder modern firm entry through specific institutional channels. To mitigate these historical "shocks," we propose the following policy interventions:

A primary policy objective should involve enhancing the inclusivity of the modern financial system within peripheral regions. Given that the contraction of grassroots finance—exemplified by Rural Credit Cooperatives—significantly obstructs firm entry, financial institutions should be encouraged to develop standardized, locally-targeted financial products. These instruments should specifically aim to decouple credit access from kinship-based informal networks. Simultaneously, the government must play a proactive role by providing credit guarantees and fiscal backing to bridge historical institutional gaps, thereby ensuring equitable financing opportunities for both external investors and local entrepreneurs.

In tandem with financial reforms, policy priorities must balance educational investment with institutional “ice-breaking” efforts. Addressing the persistent human capital deficit requires directed funding for higher and vocational education in former Tusi areas to optimize the allocation of intellectual resources. Beyond the provision of basic education, the establishment of dedicated “talent corridors” is essential to attract external professionals and elevate the regional human capital endowment. Such measures are vital for fostering a social environment that is truly conducive to modern enterprise development and professional mobility.

Furthermore, it is imperative to strengthen the integrated planning of cross-regional infrastructure to overcome historical barriers. Our mechanism and heterogeneity analyses reveal that historical spatial fragmentation—particularly within high-level Tusi jurisdictions—has led to severe failures in public goods provision. Consequently, policy efforts should focus on dismantling this governance fragmentation by prioritizing the expansion of mid-to-low-grade road networks and digital communication infrastructure. By enhancing spatial and informational connectivity, the state can effectively reduce logistics and information search costs for enterprises, leveraging regional openness to neutralize the restrictive legacy of historical boundaries.

Most importantly, our findings demonstrate that robust modern policy interventions can effectively break the negative path dependency inherited from historical legacies. Consequently, it is imperative to consolidate the achievements of poverty alleviation while sharing China's developmental experience and governance wisdom with the international community, particularly for regions still constrained by their own historical institutional burdens

7.3. Research prospects

While this study employs a rigorous empirical framework to examine the impact of the Tusi system on firm entry, several limitations remain due to data constraints and scope, pointing toward promising avenues for future inquiry

First, regarding data granularity, this research primarily utilizes county-level cross-sectional data. While our historical matching captures long-term persistence, it lacks depth in tracking the micro-dynamic characteristics of firms post-entry, such as survival quality and productivity. Future research could leverage micro-level corporate registry data to investigate whether the Tusi legacy not only suppresses the "quantity" of entry but also significantly impairs "firm longevity" or "innovation output." Such an approach would reveal the damage of institutional imprints on micro-market quality over a more comprehensive time horizon

Second, while we focus on the macro-governance legacy of the Tusi system, these historical structures are often intertwined with local ethnic culture, religious beliefs, and lineage rules. Due to the lack of fine-grained social survey data, it is challenging to completely disentangle these "bundled" informal institutions. Future studies could employ fieldwork or experimental economics to scrutinize the trust preferences, perceptions of fairness, and risk-taking behaviors of residents in former Tusi regions. By uncovering these "deep psychological traits," researchers can more accurately explain how historical institutions are "embedded" into individual decision-making, thereby shaping modern economic behavior.

Lastly, while this study focuses on the "impediment effect" of history, the "institutional repair" effect of modern policies warrants further exploration. Although our extended analysis examines the restorative role of Targeted Poverty Alleviation on firm entry, the underlying mechanisms and potential heterogeneity of this effect remain unaddressed. Future research could delve deeper into the substitution effect of formal institutions over informal ones. For instance, investigating whether digital governance and fintech can effectively penetrate deeply rooted historical social barriers would be of great value. By identifying positive factors that break historical path dependency, future studies can provide more practical theoretical foundations for institutional upgrading and social governance in peripheral regions.

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Appendix

Appendix A. figures and tables

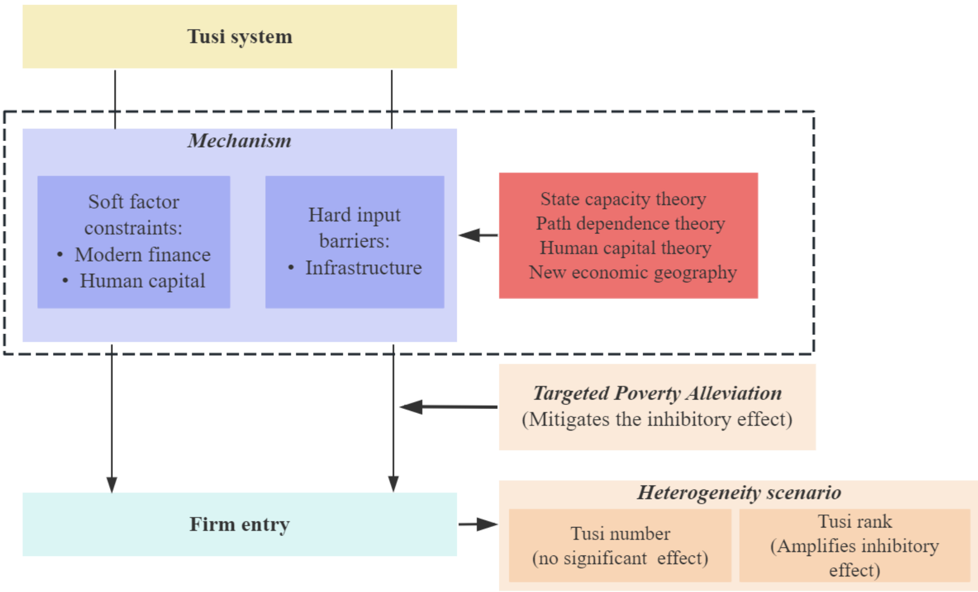

Fig. A.1. The conceptual framework

Notes: Fig. A.1 presents the framework of this paper. Drawing on classical theories such as state capacity theory, path dependence theory, human capital theory, and new economic geography, we propose mechanisms through which the Tusi system affects firm entry. Furthermore, we analyze the heterogeneous effects of this influence, particularly focusing on the number and rank of Tusi. Most interestingly, we also investigate whether modern state policy interventions—specifically Targeted Poverty Alleviation—can mitigate the negative impacts of historical extractive institutions.

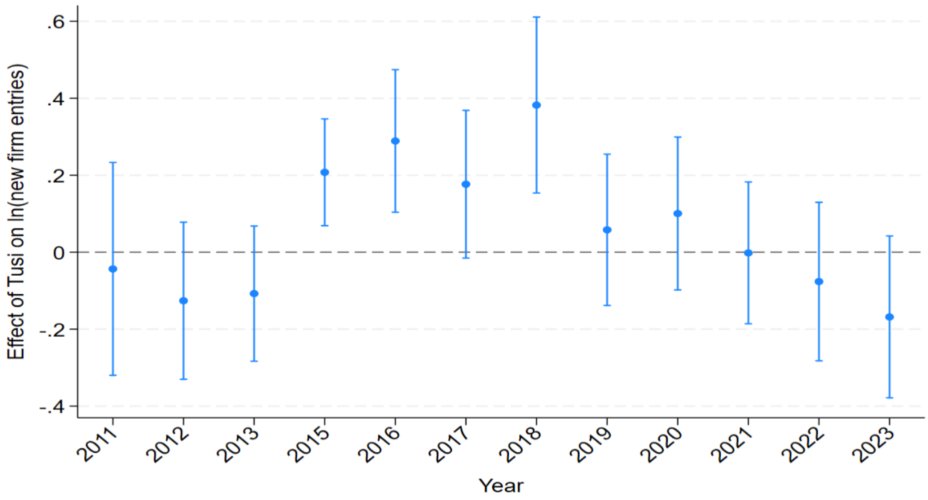

Fig. A.2. Parallel Trend Test

Notes: The consistency of our DDD estimates depends on the parallel trend assumption. This means that before the policy was implemented, the treatment and control groups should have followed the same time trend. This ensures that any unobserved factors affecting both groups remain stable over time. To test this, we use an event study approach with 2014 as the base year. The results in Fig.A.2 show that from 2011 to 2014—before the Targeted Poverty Alleviation (TPA) policy began in 2015—the estimated coefficients are not statistically significant (their 95% confidence intervals all include zero). This proves that there were no significant differences in firm entry trends between the groups before the intervention, satisfying the parallel trend assumption. After the policy was launched in 2015, the coefficients generally turned positive and became significant in several years. The trend shows an initial rise from 2015 to 2018 followed by a slight decline and stabilization between 2017 and 2023. This pattern matches the real-world logic of the policy: the massive influx of resources in the early stages led to a rapid jump in firm entry, which then leveled off as the policy became a regular part of local governance. Overall, TPA has had a significant and lasting positive effect on firm entry in regions historically affected by the Tusi system, consistent with our main findings.

Table A.1

Variable definitions.

Variables	Variable name	Definition
Panel A: Key variables		
Dependent variables	lnfirm	Log (the newly registered firms per city-year)
	firm density	the number of firms per unit of land area (number of firms / county area)
	ln private value	Log (the added value of the private economy)
Independent variables	tusi	a dummy variable representing whether a county was historically under the jurisdiction of the Tusi system (1 if yes, 0 otherwise)
Instrumental variable	wei_distance_altitude	the interaction between the log-distance from the county centroid to the nearest Ming Dynasty Wei (military garrison) and local altitude
Panel B: Mechanism variables		
	financial_outlets_density	the count of branches per unit of land area
	state_owned_bank_density	the count of branches per unit of land area
	rural_credit_cooperative_density	the count of branches per unit of land area
	primary_students_density	Log (the count of primary school students in 2020 / county area)
	middle_students_density	Log (the count of middle school students in 2020 / county area)
	college_students_density	Log (the count of college school students in 2020/ county area)
	illiterate	the illiterate of residents in the county in 2020
	ln_hospital	Log (the count of hospitals in the county in 2023)
	ln_highway	Log (highway mileage / county area)
	ln_railway	Log (the length of railway in the county in 2023 / county area)
	ln_station	Log (the count of railway stations in the county in 2023 / county area)
	ln_comm	Log (mobile phone subscribers / county area)
Panel C: Other variables		
Historical control variables	chama	a dummy variable indicating whether a county was situated along the Ancient Tea Horse Road (1 if yes, 0 otherwise)
	lnheat	the Caloric Suitability Index circa 1500 CE
Environmental control variables	temperature	a dummy variable representing whether the annual temperature is between 10-16°C (1 if yes, 0 otherwise)
	precipitation	a dummy variable representing whether the annual precipitation is between 800-1500mm (1 if yes, 0 otherwise)
Contemporary	ecoquality	a comprehensive measure of ecological suitability

control variables	industry	the secondary industrial output as a percentage of regional GDP*100
	revenue	government revenue as a percentage of regional GDP*100
	expenditure	Government expediture as a percentage of regional GDP*100
	urban	urbanization rate*100
	lnnl	Log (nighttime light brightness)

Table A.2

Heterogeneity analysis.

	(1) lnfirm	(2) lnfirm
tusi	-0.231**	0.236
	(0.116)	(0.214)
tusi_num	-0.002	
	(0.009)	
tusi_rank		-0.077**
		(0.034)
Controls	YES	YES
Time FE	YES	YES
Obs.	2,379	2,379
R-squared	0.753	0.757

Notes: This table reports the heterogeneous effects of the Tusi system on firm entry by Tusi characteristics. Num is the number of Tusi institutions in a county. Rank is coded from 1 (lowest) to 9 (highest) based on the Qing bureaucratic hierarchy, manually compiled from the Draft History of Qing and General Gazetteer of Sichuan.